JAITIN MOODALLY

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PROFILE

Computer Science student at the University of Pretoria with hands-on experience in Linux systems administration, virtualisation and real-time computer vision. Seeking internship or graduate opportunities in cyber security, networking or DevOps.

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, C#, PHP
Web & Data: Node.js, REST APIs, MySQL
Tools: Docker & Docker Compose, Git / GitHub Actions, Bash scripting, SSH & remote administration
Systems: Fedora, Debian-based distributions, Windows, Proxmox VE

EDUCATION

BSc Computer Science — University of Pretoria *2024 – Present*
Artificial Intelligence 78% • Net Centric Computer Systems 76% • Software Modelling 74% • Computer Graphics 74% • Computer Architecture 72% • Operating Systems 71%

COMPETITIONS & HACKATHONS

SANReN Cyber Security Challenge — SANReN / CSIR • National student CTF *2026 – In progress*

- Competing in the qualifier round of South Africa's national student capture-the-flag competition, working through jeopardy-style challenges across categories including web exploitation, cryptography, forensics and OSINT.

HUNT.ZA Hackathon — Universal Acceptance Bug Hunt • 2nd overall *Jun 2026*

- Placed 2nd overall at the University of Pretoria bug hunt and won the award for best email validator, judged against the other team submissions.
- Audited South African banking, government and e-commerce platforms for Universal Acceptance failures, documenting cases where valid addresses on new TLDs (.africa, .joburg, .capetown) and internationalised email addresses were silently rejected.
- Built the winning standards-based validator, handling modern TLDs and non-ASCII addresses in place of the legacy ASCII-only assumptions behind the bugs found in phase one.

Entelect University Cup — Entelect • 50th of 200 teams (top 25%) *2026*

- Solved a timed optimisation challenge in a small team, implementing a beam search over macro-actions to maximise return in the Enteloot resource economy.

PROJECTS

Gesture-Based Drone Control — Capstone Project • Team Member *2026 – Present*

- Building a system that controls a drone in real time from hand gestures, removing the need for a physical controller.
- Implemented hand tracking and gesture recognition in Python with MediaPipe, translating detected gestures into drone control commands.
- Configured automated linting and testing workflows in GitHub Actions to enforce code quality across the team repository, and contributed to system integration and testing.

Proxmox Homelab Infrastructure — Self-hosted Virtualisation Environment *2025 – Present*

- Built and maintain a Proxmox VE hypervisor running multiple VMs and LXC containers across Debian and Windows guests, managing snapshots, resource allocation and network bridge configuration.
- Deployed AdGuard Home for network-wide DNS filtering — custom upstream resolvers and blocklists, with query-log monitoring for unusual DNS activity.
- Deployed Jellyfin for on-demand media streaming across LAN and over the internet via VPN, managing user accounts, transcoding and hardware acceleration profiles.

INTERESTS

Open-source software, cyber security, home automation, artificial intelligence.